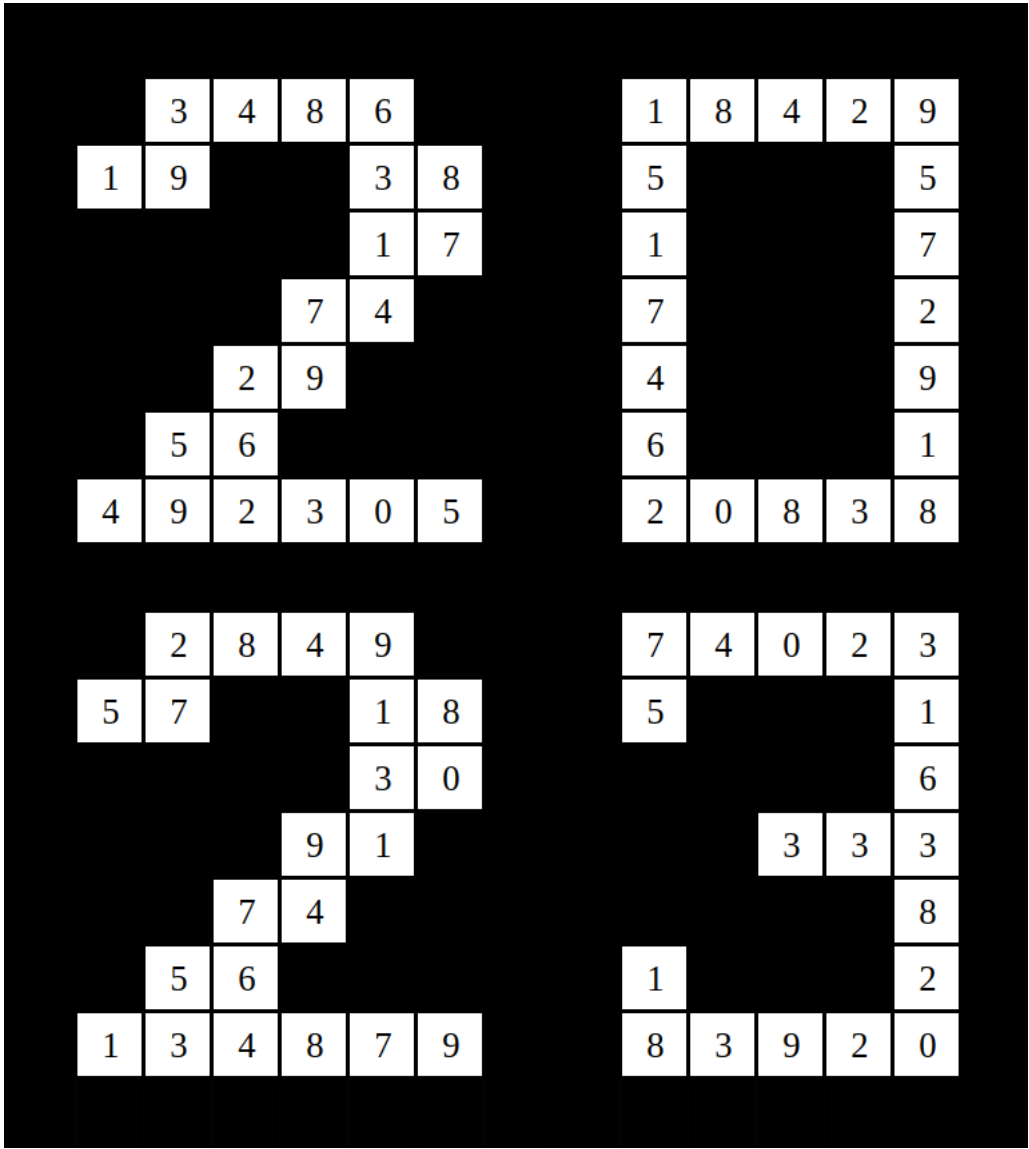


Solutions to Number Crossword

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1 Puzzle



Across

1. $(2023 \div 17) - 90)(2023 + 2628) = \mathbf{134879}$

2. $(2023 - 2020)(2023 + 4120) = \mathbf{18429}$

4. We can find the number of digits in 33^{22} using $1 + \lfloor 22 \log 33 \rfloor$, which yields 34.

Because $11|33^{22}$ and $9|33^{22}$, we can apply their divisibility rules to find the sum of the remaining values. The last digit must be 9 because the ones digit of powers of 3 cycles every 4, and dividing 22 by 4 leaves a remainder of 2. Using these digits, we can find that the current alternating sum is -4 , assuming we subtract the odd digits from the even ones, and that the current total sum is 84. There are 14 values remaining. We can express the remaining digits as $a, b, c, \dots, n$, where $A = a + b + \dots + n$ and $B = -a + b - \dots + n$. Based on the divisibility rules, we can write the following systems:

$$A \equiv 6 \pmod{9}$$

$$B \equiv 4 \pmod{11} \text{ and } B \equiv 0 \pmod{5}$$

It is also given that the average of all the remaining even numbered integers is an integer, so $b + d + \dots + n \cong 0 \pmod{7}$. So, $\frac{A+B}{2}$ must be divisible by 7. The smallest value that satisfies B is 15, so that should be the first case tried. Cycling through some values for A that work, it is seen that $A = 69$ and $B = 15$.

$$84 + 69 = 153$$

$$\frac{153-1}{8} \cdot 3$$

$$19 \cdot 3$$

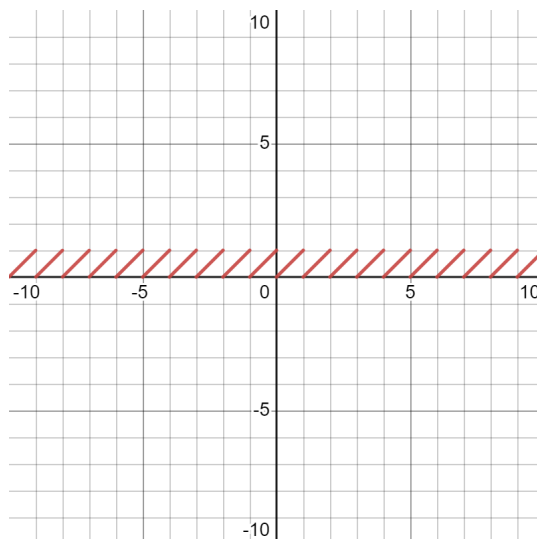
57

Credit: Patrick Was

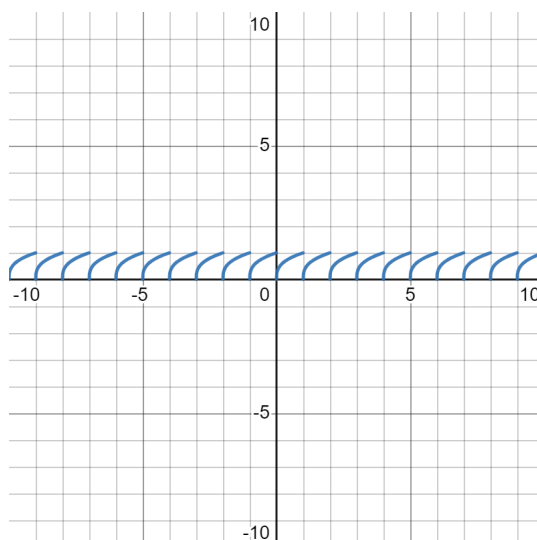
5. $(2023 \times 36) + 1195 = \mathbf{74023}$

7. $\mathbf{56} = 1 + 3 + 6 + 10 + 15 + 21$

12. A nice approach to solving integrals is by looking at the area under the curve and above the x-axis. Let's first consider the graph of $x - \lfloor x \rfloor$:



Notice that each line segment can be represented by $y = x$ where $0 \leq x \leq 1$. Now, by taking the cube root of this equation, we have $\sqrt[3]{x - \lfloor x \rfloor}$:



Again, we see that each little curve can be represented by the equation $y = x^{\frac{1}{3}}$ where $0 \leq x \leq 1$. We have $12 + 12 = 24$ of these curves, so $24 \cdot \int_0^1 \sqrt[3]{x} dx = 24 \cdot \frac{3}{4} = 18$.

Credit: Zawad Dewan

14. $333 = (3 + 3 + 3)(10 + (3 \times 3 \times 3))$

15. We can first notice that $196^2 + 578^2 = 14^4 + 4 \cdot 17^4$. Then, by Sophie Germain's Identity, we can factor the expression as $(14^2 + 2 \cdot 14 \cdot 17 + 2 \cdot 17^2)(14^2 - 2 \cdot 14 \cdot 17 + 2 \cdot 17^2) = 4 \cdot 625 \cdot 149$. This has prime factorization $2^2 \cdot 5^4 \cdot 149$, thus the number of factors is $3 \cdot 5 \cdot 2 = \mathbf{30}$.

Credit: Julia Kozak

18. $((2023 \div 119) + 20) \times 77 = \mathbf{2849}$

19. $(2023 \times 243) + 716 = \mathbf{492305}$

20. $((2023 + 75) \times 40 = \mathbf{83920}$

21. We have to represent 34 as the sum of squares. The only squares that we can use are 0, 1, 4, 6, 16, and 25. There are only two sets of sums that work: 0, 25, 9 and 16, 9, 9. In the case of 0, 25, 9, there are $3!$ ways to choose the order and $2^2 = 4$ ways to choose the sign of 5 and 3. In total there are $6 \times 4 = 24$ ways. In the case of 16, 9, 9, there are 3 ways to choose the order and $2^3 = 8$ ways to choose the sign. In total there are $3 \times 8 = 24$ ways. Adding these up, there are $24 + 24 = 48$ ways = n . Thus, $2n - 5 = 2 \times 48 - 5 = 96 - 5 = \mathbf{91}$.

Credit: Kathleen Zhang

22. $(2023 - 1940) \times 42 = \mathbf{3486}$

23. $(2023 \div 17) - 45 = \mathbf{74}$

24. In the first auction plan, both Josiah and Elie will want to bid exactly what they value the jewelry at. That way, they are guaranteed to pay less than their bid regardless of the outcome. Josiah wins here by bidding \$151. In the second auction plan, whoever has the lower bid at the end will be in danger of losing money without getting anything. This means that both bidders will continue bidding more and more to avoid losing money, even going past what they value the jewelry at. Thus Elie bids all the way up to \$137, his highest possible bid,

making Josiah's next and final bid \$138. Multiplying these two bids gives $151 \times 138 = \mathbf{20838}$.

Credit: Lee Eisenberg

25. $(2023 \div 119) + 2 = \mathbf{19}$

Down

2. $((2023 \times 375) + 106) \times 2 = \mathbf{1517462}$

3. $(2023 \times 3) + 245 = \mathbf{6314}$

5. First, the numerator is an infinite geometric series with $|r| < 1$, so we can use the formula for infinite geometric series:

$$\frac{1}{1 - .99} = \frac{1}{.01} = 100$$

Let's try to manipulate the summation in the denominator. We can factor the denominators and notice a potential partial fraction decomposition:

$$\sum_{k=1}^n \left(\frac{2k+2}{k(k+2)} - \frac{2k+4}{(k+1)(k+3)} \right)$$

We want to factor the first term into some $\frac{A}{k} + \frac{B}{k+2}$ and the second term into some $\frac{C}{k+1} + \frac{D}{k+3}$. Combining each pair of fractions we get $\frac{A(k+2)+Bk}{k(k+2)}$ and $\frac{C(k+1)+D(k+3)}{(k+1)(k+3)}$. We want to match up the coefficients of k and the constants, so we can see that $A = 1$ due to the constant term so $B = 1$ to get a $2k$. We solve $C + D = 2$ and $C + 3D = 4$ to get that $C = 1$ and $D = 1$ as well. So, this simplifies to

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} + \frac{1}{k+2} - \frac{1}{k+3} \right)$$

We can see that this telescopes. Let's look at the first few terms:

$$\begin{aligned} &1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \\ &\quad \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} \\ &\quad \quad \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} \\ &\quad \quad \quad \frac{1}{4} - \frac{1}{5} + \frac{1}{6} - \frac{1}{7} \end{aligned}$$

And so on. All terms after $\frac{1}{4}$ will completely cancel since there will be 4 instances of each with alternating signs. All we have left are 1 , $\frac{1}{2}$, and $\frac{1}{3}$. The $\frac{1}{2}$ cancels, and we are left with $1 + \frac{1}{3} = \frac{4}{3}$. The sum is equal to $\frac{4}{3}$, so

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n (.99)^k}{\sum_{k=1}^n \left(\frac{2k+2}{k^2+2k} - \frac{2k+4}{k^2+4k+3} \right)} = \frac{100}{\frac{4}{3}} = 100 \frac{3}{4} = \mathbf{75}.$$

Credit: Ethan Sharma

6. $(2023 \div 119) + 70 = \mathbf{87}$

8. $(2023 \times 4) + (20230 \div 17) - 151 = \mathbf{9131}$

9. Since the number raised to the fifth power is positive, we know that our only options are $1^5 = 1$ or $2^5 = 32$. $3^5 = 243$ is too large.

Let's say we choose 1^5 , which is an odd number. For the sum to be prime, our only possible scenario is $1^5 + 2 = 3$. We can be fairly certain, though, that the answer is bigger than 3. Let's try using 2^5 to get bigger solutions.

$2^5 = 32$. Starting from the largest prime less than 100 and going down:

$$97 - 32 = 65$$

$$89 - 32 = 57$$

$$83 - 32 = 51$$

$$79 - 32 = 47$$

47 is prime. Thus, our answer is **79**.

Credit: Daphne Qin

10. It's reasonable to infer that the terms are supposed to somehow cancel each other out. Therefore, we should actively be looking for a way to rewrite each term in this sequence as a difference between two expressions.

Notice that the sequence N can be rewritten as

$$\frac{2}{2(2^2 - 1)} + \frac{2}{3(3^2 - 1)} + \frac{2}{4(4^2 - 1)} + \frac{2}{5(5^2 - 1)} + \dots$$

in which every term can be represented as

$$\frac{2}{a(a^2 - 1)}$$

for some number a. This expression can then be written as the following:

$$\frac{2}{a(a+1)(a-1)}$$

Notice that $(a+1) - (a-1) = 2$. Therefore, we can rewrite the expression as:

$$\frac{(a+1) - (a-1)}{a(a+1)(a-1)} = \frac{(a+1)}{a(a+1)(a-1)} - \frac{(a-1)}{a(a+1)(a-1)}$$

$$= \frac{1}{(a-1) \times a} - \frac{1}{a \times (a+1)}$$

So, the sequence N can be rewritten as

$$\left(\frac{1}{1 \times 2} - \frac{1}{2 \times 3} \right) + \left(\frac{1}{2 \times 3} - \frac{1}{3 \times 4} \right) + \left(\frac{1}{3 \times 4} - \frac{1}{4 \times 5} \right) + \dots$$

It's important to note that for any n th term of the sequence, the a value is not equal to n but $n + 1$. The sum of the first n th terms of the sequence N can thus be written as

$$\frac{1}{1 \times 2} - \frac{1}{(n+1) \times (n+2)} = \frac{189}{380}$$

$$\frac{1}{(n+1) \times (n+2)} = \frac{1}{380}$$

The value of n is **18**.

Credit: Lucia Liu

18. Start by listing out the first few primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29... Now, we can start by testing the smallest two-digit integer and go up from there.
 There are four primes less than 10 and 11. 4 divides neither.
 There are five primes less than 12 and 13. 5 divides neither.
 There are six primes less than 14, 15, 16, and 17. 6 divides none of them.
 There are seven primes less than 18 and 19. 7 divides neither.
 There are eight primes less than 20, 21, and 22. 8 divides none of them.
 Now there are nine primes less than 23, 24, 25, 26 and 27, and 28. 9 divides **27**.

Credit: Mr. Kats